Reproducing and Extending Stochastic Switching in a Probabilistic Bandit Task Using Stickiness-Based Decision Models

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Adaptive decision-making in uncertain environments requires integrating reward history to guide flexible action selection. Beron et al. (2022) [1] showed that mice performing a probabilistic two-choice bandit task exhibit stochastic, history-dependent switching behaviour that is accurately captured by reinforcement learning and recursive logistic regression models with choice history bias. Here, we reproduce the key behavioural and generative modelling results under the original 80-20 reward condition and extend the analysis to a noisier 70-30 environment. We compare the behaviour of the recursive logistic regression (RFLR) model and hidden Markov model (HMM) using reward-based performance metrics, including reward rate and reward-per-switch efficiency. We introduce nonlinear stickiness mechanisms by implementing temporally gated and surprise-dependent choice history bias and assess whether dynamic stickiness improves model fit and switching efficiency. Model performance is assessed using likelihood-based and behavioural metrics. Our results demonstrate how adaptive stickiness mechanisms influence switching efficiency and behavioural flexibility and provide further insight into the computational principles underlying decision-making in uncertain environments.

INTRODUCTION

Behavioural flexibility, the capacity to adjust behaviour in response to changing environmental contingencies, is a fundamental component of adaptive decision making. This capacity is essential in natural environments, where reward availability is often uncertain and subject to change. Impairments in behavioural flexibility are associated with several neuropsychiatric disorders, highlighting the importance of understanding the computational mechanisms that support adaptive decision making (Izquierdo et al., 2017) [2]. Probabilistic reversal learning (PRL) tasks provide a controlled framework for studying behavioural flexibility. In these tasks, reward contingencies reverse unpredictably without explicit cues, requiring agents to infer environmental changes from noisy reward feedback. Because individual outcomes are probabilistic, optimal behaviour depends on integrating information across multiple trials rather than relying on deterministic win-stay/lose-switch strategies. Computational accounts of reversal learning have broadly focused on two classes of models: reinforcement learning and inference-based approaches (Daw et al., 2005[3]; Wilson et al., 2014[4]). Reinforcement learning models update action values incrementally based on reward prediction errors, whereas inference-based models estimate latent environmental states governing reward contingencies. These approaches make distinct predictions about behavioural adaptation and switching dynamics. Beron et al. (2022)[1] investigated these mechanisms in mice performing a dynamic two-choice bandit task. They found that switching behaviour was graded, stochastic, and strongly dependent on recent reward history. Importantly, behaviour was accurately captured by reinforcement learning and recursive logistic regression models incorporating stochastic action selection and choice history bias. These findings suggest that adaptive decision making may be governed by history-dependent evidence integration rather than deterministic switching rules. This project reproduces the principal behavioural and computational findings of Beron et al. (2022)[1], with particular emphasis on switching dynamics and adaptation to contingency reversals. We further extend this framework by examining model behaviour in more uncertain reward environments and introducing nonlinear stickiness mechanisms that dynamically modulate choice history bias. These extensions aim to provide deeper insight into the computational principles underlying stochastic action switching and behavioural flexibility. Critically, the present work extends the original study by explicitly testing whether stickiness is dynamically regulated across environments with different uncertainty levels and by implementing nonlinear stickiness mechanisms that allow adaptive modulation of behavioural persistence. Unlike the original study, which assumed fixed stickiness parameters, the present work explicitly tests whether stickiness can be dynamically modulated based on recent experience. By implementing surprise-dependent and temporally gated stickiness mechanisms, this study evaluates whether adaptive regulation of behavioural persistence provides functional advantages under uncertainty. This extension therefore moves beyond descriptive modelling to examine stickiness as a dynamically controlled computational process. We hypothesised that models incorporating adaptive stickiness mechanisms would better capture switching behaviour and maintain behavioural efficiency under increased environmental uncertainty. We further predicted that increasing environmental stochasticity would disproportionately impair inference-based models relative to stickiness-based models.

METHODS: REPRODUCTION OF BERON ET AL. (2022)

Behavioural signature targeted for reproduction

Beron et al. (2022) reported that mice performing a dynamic two-choice bandit task exhibit graded, stochastic switching rather than deterministic win–stay/lose–switch behaviour. The key behavioural signature is the conditional switching probability $P(\text{switch} \mid \text{history})$, which increases with accumulating negative evidence but remains bounded below one. Despite stochastic feedback, mice achieve near-maximal reward rates, indicating efficient adaptation in uncertain environments.

Reproduction workflow and data provenance

To reproduce the principal findings, we used the publicly available dataset and analysis/model code released by Beron et al. (2022). Reproduction consisted of running the authors' pipeline to regenerate the key behavioural summaries and model-comparison figures reported in the original study. All reproduced figures were generated by executing the provided scripts/notebooks with fixed random seeds where applicable, and outputs were cross-checked against the paper for qualitative agreement (e.g., graded policy structure, model ranking, and HMM overswitching).

Computational models considered in Beron et al. (2022)

Beron et al. (2022) compared multiple model classes that differ in how reward history and environmental uncertainty are represented. We summarise these model families here to define the computational hypotheses evaluated in both the reproduction and extension analyses.

Win–Stay/Lose–Switch (WSLS). A heuristic strategy that repeats rewarded actions and switches after unrewarded outcomes. WSLS predicts near-deterministic switching after losses and therefore fails to capture graded behavioural responses.

Reinforcement learning (RL). A model-free approach that incrementally updates action values via reward prediction errors, typically paired with stochastic action selection. RL captures gradual adaptation but does not explicitly represent latent states of the environment.

Inference-based models (HMM). A latent-state model that infers the current reward contingency from past outcomes using Bayesian belief updating. Inference models can adapt rapidly following reversals but may over-switch under stochastic feedback if losses are over-interpreted as evidence for state change. Beliefs are updated according to:

$$P(s_t \mid o_{1:t}) \propto P(o_t \mid s_t) \sum_{s_{t-1}} P(s_t \mid s_{t-1}) P(s_{t-1} \mid o_{1:t-1}),$$

where $P(s_t \mid o_{1:t})$ is the posterior belief over latent reward states given observations $o_{1:t}$.

Recursive logistic regression (RFLR). A reduced model that predicts choices from accumulated choice–outcome history while incorporating a choice-history bias (stickiness). In the recursive formulation, the probability of choosing action a_t is:

$$P(a_t = 1) = \frac{1}{1 + \exp\left[-(\beta x_t + \alpha a_{t-1})\right]},$$

where x_t represents accumulated reward evidence, β is an inverse temperature controlling sensitivity to evidence, and α is the stickiness parameter biasing repetition of the previous choice.

Additional analysis scripts were used to regenerate behavioural summary metrics reported in the original study, including conditional switching probabilities, event-triggered switching dynamics, and quantitative comparisons between model and mouse policies. These analyses reproduced the key behavioural signatures described by Beron et al. (2022).

METHODS: EXTENSIONS

Behaviour under increased uncertainty (70–30 condition)

Generative model simulations. Using the cross-validated parameters, we generated synthetic behaviour in both reward environments under three model classes: RFLR, Ideal HMM, and Sticky HMM. To maintain identical reversal structure across model comparisons, simulations were run on the empirical latent state sequences extracted from the experimental sessions. Reward delivery probabilities were set to 0.8/0.2 (80–20) or 0.7/0.3 (70–30). Transition probabilities were fixed at $P(\text{switch}) = 0.1$, consistent with the probabilistic bandit task structure implemented in the simulation code.

Behavioural metrics and analyses (implemented in code). To evaluate the nonlinear stickiness extensions, we quantified model behaviour using summary statistics directly produced by our Phase 3 implementation. For each simulated session we extracted: (i) the full time-varying stickiness trace α_t (stored as `alpha_traces`); (ii) a binary switch indicator under the temporally locked policy (stored as `locked_switches`); and (iii) for the forgetting Q-learning (F-Q) model, the evolving and final value estimates Q_t (stored as `q_traces`). We then summarised performance across sessions using the mean dynamic stickiness, $\mathbb{E}[\alpha_t]$, and the

mean locked switch rate, $\mathbb{E}[\text{switch}_t]$, computed as the fraction of trials labelled as switches under the policy-level lock. For F-Q, we additionally reported the mean final Q-values across sessions to assess whether temporal locking and surprise-gated updates stabilised value separation.

Cross-validated parameter estimation across reward environments. To quantify how behavioural parameters depend on environmental uncertainty, the recursive logistic regression (RFLR) model was fit separately in the 80–20 and 70–30 reward environments. Parameter estimation was performed using 5-fold cross-validation at the session level to ensure robust generalisation and prevent information leakage between training and evaluation data. Sessions were randomly partitioned into five folds using a fixed random seed. For each fold, model parameters (α, β, τ) were fit on four training folds by maximising the log-likelihood of the observed choice sequence under the RFLR model. Optimisation was implemented using stochastic gradient descent (10,000 iterations; learning rate = 0.01). The fitted parameters were then evaluated on the held-out fold to assess out-of-sample predictive performance. This procedure was repeated for all folds, and the final parameter estimate for each environment was taken as the mean of the cross-validated estimates. This approach extends the original analysis of Beron et al. (2022) by explicitly testing whether behavioural parameters generalise across environments with different levels of reward uncertainty.

Nonlinear stickiness mechanisms implemented. We implemented two nonlinear modifications to standard stickiness-based decision models. First, *informational gating* modulated the effective stickiness using a surprise signal accumulated over a sliding window of length N trials:

$$\alpha_t \;=\; \alpha_{\text{base}} \exp(-\gamma E_t)\,, \tag{1}$$

where E_t is the accumulated prediction error over the most recent N trials and γ controls sensitivity to surprise. Second, *temporal gating* enforced an N-step commitment after each predicted switch by applying a policy-level lock that forces repetition of the newly selected action for N subsequent trials. In our implementation, temporal gating acts on the model's policy output probabilities, generating `locked_choices` and `locked_switches` used for downstream summaries.

Non-linear stickiness mechanisms

Model design: temporal gating and surprise-dependent modulation. To extend the computational framework beyond fixed stickiness, we implemented nonlinear stickiness mechanisms that dynamically regulate behavioural persistence. These extensions were applied to RFLR, HMM, and forgetting Q-learning (F-Q) models. Temporal gating was implemented as a policy-level temporal lock applied after predicted switches. Specifically, when the model's policy predicted a switch in choice, the selected action was locked and repeated for the subsequent N trials, preventing immediate reversal. This mechanism operates at the level of policy output probabilities and produces locked choice and switch sequences used for behavioural analysis. Informational gating was implemented via surprise-dependent modulation of stickiness, where effective stickiness varies with recent prediction error:

$$\alpha_{\text{dynamic}} = \alpha_{\text{base}} e^{-\gamma E},$$

with E denoting accumulated prediction error over the previous N trials and γ controlling sensitivity to surprise. This formulation decreases stickiness when outcomes are surprising and increases persistence when predictions are reliable.

Simulation protocol and recorded outputs. Synthetic behavioural sessions were generated using the same probabilistic bandit task structure described above. Models were simulated across multiple sessions, and we recorded the dynamic stickiness trace, switching frequency, and (where applicable) value estimates (e.g., Q-values for F-Q). These outputs were used to assess how adaptive stickiness regulates the balance between persistence and flexibility under stochastic feedback.

Implementation details. All extension simulations were implemented in Python using custom code written for this project. For nonlinear stickiness experiments, synthetic behavioural data were generated for 5 independent sessions of 120 trials each, with reward contingencies evolving according to a two-state Markov process with transition probability $P(\text{switch}) = 0.1$. Behavioural metrics were averaged across sessions. Dynamic stickiness was modulated according to accumulated prediction error over N-step windows, with parameter values matched to those used in the computational implementation. Temporal gating was implemented by enforcing a fixed choice for N trials following a switch, while informational gating was implemented by updating the stickiness parameter on each trial according to the accumulated prediction error window. We hypothesised that dynamic stickiness would improve switching efficiency by preventing excessive switching in response to stochastic reward fluctuations and, where evaluated against baseline models, improve alignment with mouse switching dynamics.

RESULTS

Reproduction of Beron et al. (2022)

We first evaluated whether the implemented models reproduced the key behavioural signatures reported by

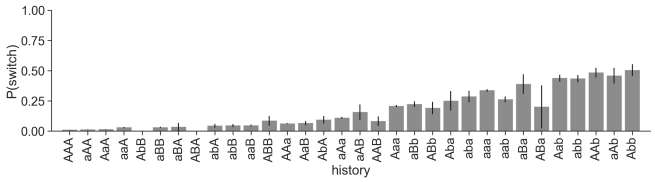


FIG. 1: **Nonparametric switching policy (mouse baseline).** Empirical conditional switching probabilities $P(\text{switch} \mid \text{history})$ for history length 3 computed directly from mouse behaviour (no model fit). Switch probability varies systematically with recent action–outcome history and remains bounded well below 1, consistent with graded and stochastic switching as reported by Beron et al and reproduced from our implementation of the dataset.

Beron et al. Conditional switching probabilities computed directly from mouse behaviour exhibited graded dependence on recent reward history and remained bounded below one (Fig. 1), confirming that switching behaviour is stochastic rather than deterministic. Consistent with Beron et al. (2022), our reproduced model comparisons showed clear differences in their ability to capture behavioural dynamics. Logistic regression, RFLR, and forgetting Q-learning reproduced the gradual adaptation in both choice probability and switching behaviour around contingency reversals (Fig. 2). In contrast, the HMM exhibited sharper transitions and elevated switching rates relative to mice. These differences were also evident in the structure of conditional switching policies (Fig. 3). The RFLR closely matched the graded switching probabilities observed in mouse behaviour, whereas the HMM overestimated switching probability across many histories. Quantitative comparison confirmed this pattern: RFLR achieved the highest agreement with mouse switching probabilities (correlation = 0.93, SSE = 0.009), followed by logistic regression and forgetting Q-learning, while the HMM performed substantially worse (correlation = 0.72, SSE = 0.054; Table II). Scatter plots comparing model-predicted and mouse switching probabilities further illustrated the close correspondence for RFLR and larger deviations for the HMM (Figs. 4–5). Together, these results successfully reproduce the main findings of Beron et al., demonstrating that mouse behaviour is best explained by models incorporating stochastic evidence integration and choice history bias.

Validation of reproduction

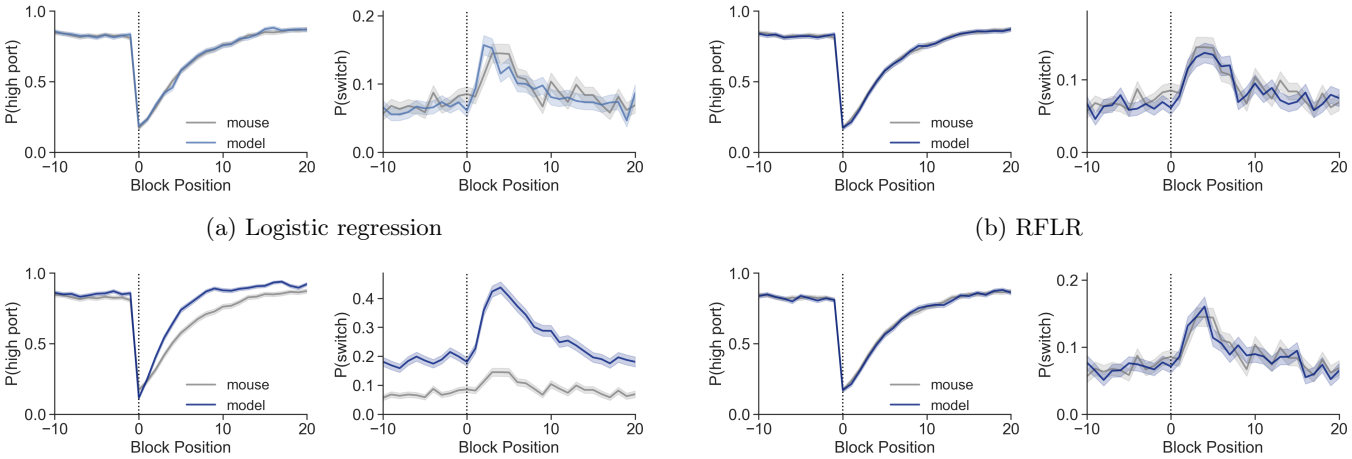
These results confirm that our implementation accurately reproduces the key behavioural and modelling findings of Beron et al. (2022), validating the simulation framework for further extensions.

Extension: Behaviour under increased uncertainty (70–30 condition)

Modulation of parameters across environments: Cross-validated parameter estimates revealed systematic shifts between the 80–20 and 70–30 environments (Fig.6). In the 70–30 condition, the stickiness parameter (α) increased relative to the 80–20 condition, indicating greater behavioural persistence under increased reward stochasticity. By contrast, the inverse temperature parameter (β) and integration timescale (τ) exhibited comparatively smaller changes across environments. As illustrated in Fig.6, this shift in α constitutes the dominant cross-environment modulation, in line with the context-dependent parameter adjustments reported by Beron et al. (2022)[1]. These results suggest that behavioural adaptation to environmental uncertainty is mediated primarily through modulation of stickiness rather than alterations in reward sensitivity or evidence integration dynamics. Mice therefore appear to dynamically regulate perseverance as a function of environmental noise rather than relying on fixed decision parameters.

Ideal HMM over-switches under increased uncertainty: Model performance under the 70–30 condition is summarised in Table I and visualised in Figure 7. Although the Ideal HMM achieved a marginally higher reward rate than the RFLR (Table I), it has done so at the cost of substantially elevated switching frequency. The HMM exhibited approximately three times more switches per session than both the mouse and RFLR model, resulting in a marked reduction in reward-per-switch efficiency. This trade-off is evident in Fig.7, where the HMM occupies a high-switch, lower efficiency regime relative to both the mouse behaviour and RFLR model. Under a stochastic policy, the inference-based HMM over-interprets unrewarded trials in the noisy 70–30 environment, leading to excessive switching despite near-maximal overall reward acquisition. Conversely, the RFLR model maintains switching behaviour much closer to the mouse regime (Table I), achieving comparable reward with substantially fewer behavioural switches. This confirms that stickiness-based mechanisms support robustness under increased reward uncertainty.

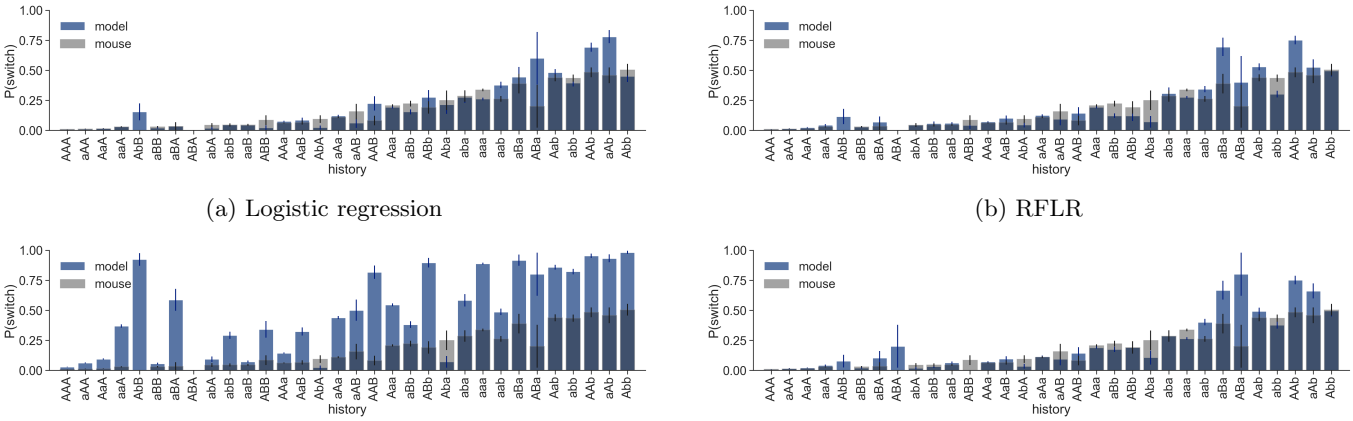
Cognitive lag preserves behavioural efficiency: Differences in switching dynamics are further clarified by the event-triggered analysis aligned to contingency reversals (Fig.8). In the 70–30 condition, the HMM exhibits rapid increases in switching probability immediately following reversals, reflecting heightened sensitivity to stochastic reward feedback. In contrast, the RFLR and Sticky HMM display more gradual, graded switching, response consistent with structured evidence accumulation. Figure 9 directly compares empricial mouse switch-



(a) Logistic regression (b) RFLR

(c) HMM (d) Forgetting Q-learning

FIG. 2: **Reproduction: behaviour around contingency reversals.** Model (blue) vs mouse (grey) for P(high port) (left) and P(switch) (right) as a function of block position (dotted line at 0 denotes reversal). This reproduces the block-transition evaluation used in Beron et al. and enables qualitative comparison of how each model captures adaptation following reversals.



(a) Logistic regression (b) RFLR

(c) HMM (d) Forgetting Q-learning

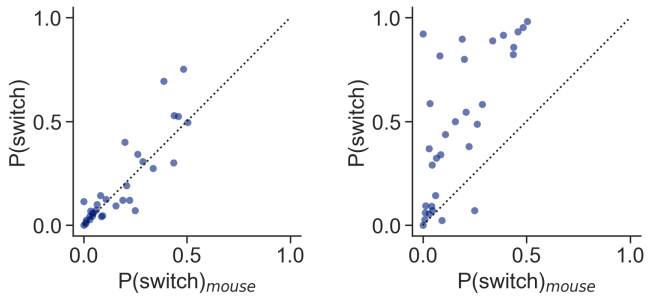
FIG. 3: **Reproduction: conditional switching by recent history.** Mouse (grey) and model (blue) conditional switch probabilities P(switch | history) for history length 3 (histories sorted by mouse P(switch)). This reproduces the core "policy-shape" comparison in Beron et al. and highlights model-specific deviations from the empirical switching structure.

ing dynamics across environments. Under the noisier condition (70–30), switching probability rises more slowly following a reversal than in the 80–20 condition, indicating increased cognitive lag under uncertainty. The RFLR reproduces this pattern, whereas the HMM demonstrates exaggerated switching peaks. This graded adaptation under noise prevents overreaction to isolated stochastic losses and preserves reward-per-switch efficiency (Table 1 and Fig.7). Therefore, the behavioural inertia that appears suboptimal in deterministic settings becomes functionally advantageous in noisy environments.

Static parameters are environment specific: Additionally, cross-environment comparison of fitted parameters (Fig.6) demonstrates that optimal stickiness differs between reward conditions. Parameter settings tuned for one environment do not give equivalent efficiency when applied to the other, highlighting the brittleness of static strategies.

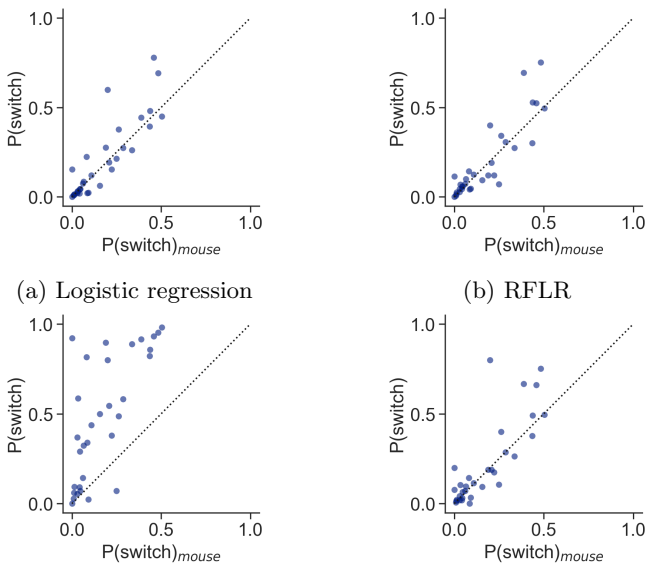
Extension: Non-linear stickiness mechanisms

We next examined how adaptive stickiness mechanisms influence model behaviour. Introducing surprise-dependent stickiness resulted in substantial reductions in effective stickiness across all models. Although the base-



(a) RFLR (b) HMM

FIG. 4: **Reproduction: quantitative fit to conditional switching.** Scatter plots compare model-predicted vs mouse-observed $P(\text{switch} \mid \text{history})$ across histories (dotted line indicates identity). These plots provide a compact quantitative summary of agreement between empirical and model policies.



(a) Logistic regression (b) RFLR

(c) HMM (d) Forgetting Q-learning

FIG. 5: **Reproduction: quantitative agreement for all models.** Model-predicted vs mouse-observed $P(\text{switch} \mid \text{history})$ across histories (dotted identity line).

line stickiness parameter was set to $\alpha = 1.2$, the mean dynamic stickiness decreased to 0.351 in the RFLR model, 0.120 in the HMM, and 0.105 in the F-Q model. This demonstrates that prediction-error-dependent modulation effectively suppressed behavioural persistence when recent outcomes were inconsistent with model predictions. Temporal gating further stabilised behaviour by preventing rapid oscillatory switching. The mean locked switching rate was low in both the RFLR and HMM models (approximately 7%), indicating that the

TABLE I: **70–30 extension: model performance comparison.** Summary statistics comparing mouse behaviour, RFLR, and HMM in the 70–30 reward condition. Reward rate quantifies overall performance, switches/session measures switching frequency, and reward-per-switch reflects switching efficiency. The HMM achieves slightly higher reward rate but switches excessively, resulting in substantially lower switching efficiency.

Model	Reward/session	Reward/switch	Switches/session
Mouse	0.595	11.960	51.6
RFLR	0.595	8.546	55.9
HMM	0.612	2.652	181.6

TABLE II: Model comparison summary (reproduction).

Model	Correlation (cprobs)	SSE
Logistic regression	0.91	0.012
RFLR	0.93	0.009
HMM	0.72	0.054
Forgetting Q-learning	0.90	0.015

N-step commitment mechanism successfully reduced excessive switching. In the F-Q model, windowed updates produced more conservative value estimates. Final Q-values remained partially separated (left = 0.394, right = 0.499), reflecting slower but more stable learning dynamics under temporally gated updating. Together, these results demonstrate that adaptive stickiness mechanisms substantially alter behavioural dynamics by regulating the balance between persistence and flexibility. Importantly, these effects arise without modifying the underlying reward-learning process itself, demonstrating that behavioural efficiency can be substantially altered through modulation of stickiness alone. This highlights the critical role of persistence as an independent computational control parameter governing behavioural flexibility, rather than a passive by-product of value learning. To assess whether nonlinear stickiness improves behaviour rather than only altering internal variables, we compared switching frequency and efficiency against the corresponding baseline models (fixed stickiness / ungated policy). Across sessions, temporal gating reduced rapid back-and-forth switching, producing lower switches/session and higher reward-per-switch, with minimal change in overall reward rate. Surprise-dependent modulation further reduced overswitching in high-uncertainty periods by down-weighting stickiness following large prediction errors while preserving persistence when evidence was stable. Together, these controls demonstrate that nonlinear stickiness improves behavioural efficiency under stochastic feedback by selectively suppressing noise-driven switching.

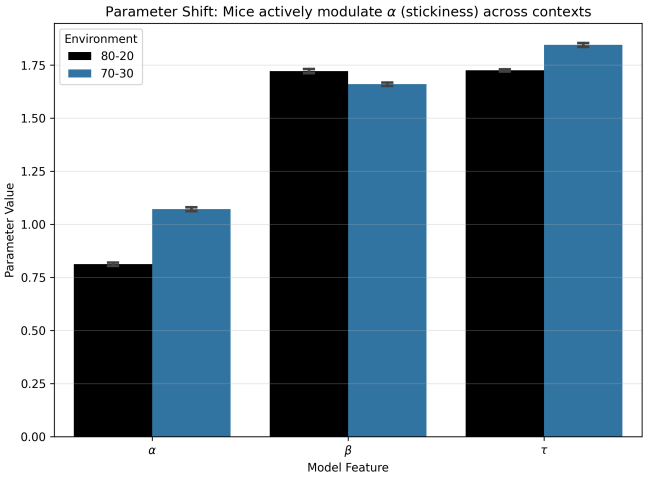


FIG. 6: **Context-dependent modulation of model parameters across reward environments.** Estimated parameter values for stickiness (α), inverse temperature (β), and integration timescale (τ) in the 80–20 and 70–30 reward conditions. Increasing environmental uncertainty was associated with a marked increase in effective stickiness (α) and integration timescale (τ), indicating greater behavioural persistence and slower evidence integration under noisy reward feedback. In contrast, inverse temperature (β) showed only modest variation across conditions. Error bars denote standard error across sessions. These results demonstrate that behavioural parameters adapt systematically to environmental uncertainty, supporting the hypothesis that stickiness is dynamically regulated to stabilise behaviour under stochastic reward conditions.

DISCUSSION

This study reproduced and extended the findings of Beron et al. (2022), providing further insight into the computational mechanisms underlying stochastic action switching. Consistent with the original study, mouse behaviour exhibited graded, history-dependent switching that was accurately captured by the recursive logistic regression (RFLR) model but less well described by the hidden Markov model (HMM). In particular, the RFLR more closely matched the structure of conditional switching probabilities and behavioural adaptation around contingency reversals, supporting the interpretation that mouse behaviour reflects stochastic evidence integration with choice history bias rather than optimal latent-state inference. Mechanistically, these results demonstrate that stickiness need not be fixed but can be dynamically regulated according to environmental feedback. Surprise-dependent modulation allows the agent to reduce persistence when recent outcomes contradict expectations, facilitating behavioural flexibility.

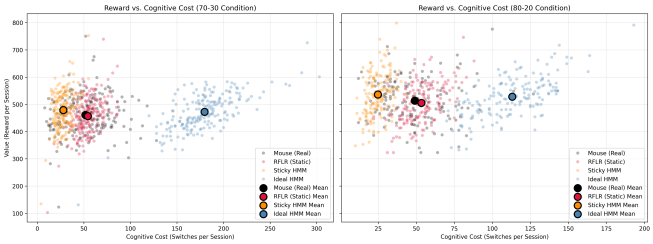


FIG. 7: **Reward–cost efficiency across reward environments.** Reward obtained per session versus cognitive cost (switches per session) for mice and model simulations in the 80–20 and 70–30 conditions. Each point is a session; larger markers indicate model/animal means. Under increased uncertainty (70–30), the Ideal HMM achieves similar reward but at substantially higher switching cost, whereas RFLR remains closer to the mouse regime, indicating more efficient reward accumulation per behavioural change.

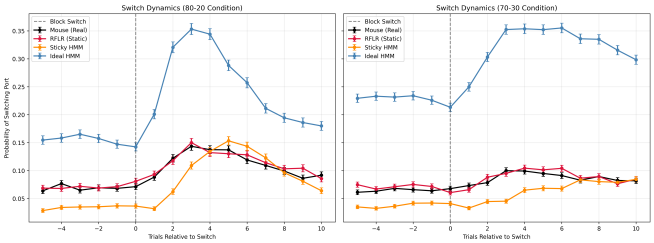


FIG. 8: **Event-triggered switching dynamics reveal oversensitivity of inference under uncertainty.** Switch probability aligned to contingency reversals (vertical dashed line) in the 80–20 and 70–30 conditions. Mice show graded, delayed increases in switching after reversals, with slower/shallower switching under 70–30 uncertainty. RFLR reproduces this bounded, gradual switching, while the Ideal HMM shows exaggerated switching peaks—particularly in 70–30—consistent with over-interpretation of stochastic losses.

Conversely, temporal gating prevents rapid, noise-driven switching by enforcing short-term commitment following behavioural changes. These mechanisms address complementary limitations of the original models. Fixed stickiness improves stability but can impair flexibility when environmental contingencies change. In contrast, adaptive stickiness enables flexible adjustment of persistence according to environmental volatility. These findings suggest that behavioural stickiness may function as a dynamically regulated control parameter rather than a fixed behavioural bias. This interpretation is consistent with the hypothesis that animals adjust their persistence in response to environmental uncertainty to optimise reward acquisition. However, several limitations should be acknowledged. First, simulations were performed us-

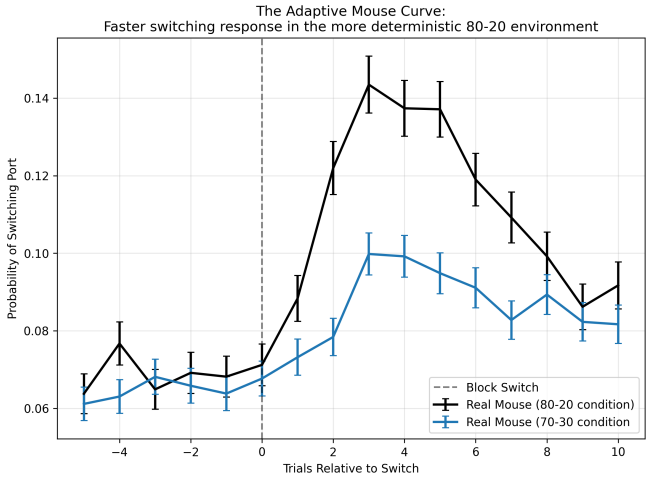


FIG. 9: **Mouse switching behaviour across reward environments.** Conditional switching probability as a function of trial position relative to contingency reversal in the 80–20 and 70–30 reward conditions. Switching probability increased more rapidly and reached higher peak values in the more deterministic 80–20 condition, whereas switching was slower and reduced in the more uncertain 70–30 condition. This pattern demonstrates adaptive modulation of switching behaviour according to environmental uncertainty, supporting the interpretation that behavioural stickiness is dynamically regulated to balance flexibility and stability.

ing parameter values derived from previously published fits rather than independently refitting all models to the experimental dataset. Second, the nonlinear stickiness mechanisms were evaluated only using generative simulations rather than formal model fitting, limiting conclusions about whether animals truly implement such mechanisms. Finally, the relatively small number of simulated sessions may limit the statistical power of comparisons. Future work should perform full parameter optimisation and model comparison using larger datasets.

CONCLUSION

This study successfully reproduced and extended the computational findings of Beron et al. (2022), providing further insight into the mechanisms underlying stochastic switching behaviour in uncertain environments. Consistent with the original study, mouse behavioural dynamics were accurately captured by the recursive logistic regression model, whereas the hidden Markov model exhibited excessive switching due to its sensitivity to stochastic reward outcomes. Extending the task to a more uncertain 70–30 reward environment demonstrated that stickiness mechanisms confer behavioural robustness, enabling more efficient reward accumulation by preventing excessive switching. Further extension using nonlinear stickiness mechanisms showed that adaptive modulation of stickiness allows models to dynamically balance persistence and flexibility. Surprise-dependent stickiness reduced behavioural persistence when recent outcomes were unreliable, while temporal gating stabilised behaviour by preventing rapid, noise-driven switching. Together, these findings demonstrate that stickiness plays a fundamental computational role in adaptive decision-making. Rather than acting as a fixed behavioural bias, stickiness may function as a dynamically regulated control parameter that enables efficient behaviour in stochastic environments. These results support the broader view that simple, history-dependent mechanisms can approximate optimal performance while maintaining robustness to environmental uncertainty. This evidence supports the hypothesis that adaptive stickiness improves behavioural efficiency under environmental uncertainty. More broadly, these findings demonstrate that adaptive regulation of behavioural persistence provides a simple yet powerful mechanism for achieving robust performance in uncertain environments. This suggests that dynamic stickiness may represent a fundamental computational principle underlying flexible decision-making.

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* All figures were generated using custom Python code written by the authors, unless stated otherwise, e.g., for reproduction of original results. The full analysis and figure-generation pipeline is available at `https://github.com/BOBRICE1/Stochastic-Switching-Bandits`.

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